MATH DOMINO STUDIO & 12-FRAME REGROUPING LAB

Triangular Combinatorics, Subitizing & Part-Whole Decompositions

CARD 02 / 10 • DOMI-01

MATHEMATICAL REFERENCE

STUDIO CONTINUUM: 1·Ten-Frames | 2·Dominoes | 3·Clock | 4·Fractions | 5·Primes | 6·Algebra | 7·Probability | 8·Volvelle | 9·Quest | 10·Rack | 11·SC-32

1. WHAT IS THIS MANIPULATIVE?

TRIANGULAR COMBINATORICS & PART-WHOLE PAIRS

Double-six dominoes form a triangular math system.
It contains all unordered pairs [a|b] from 0 to 6:
Total Tiles = T(7) = (7 × 8) / 2 = 28 Unique Tiles
Each tile is a 2-part additive partition of 0–12!

MODULAR 28.0 × 14.0 MM TILE DATUM

- Tile Size: 28.00 × 14.00 × 7.00 mm (4U×2U×1U)
- Tactile Pips: 3.20 mm ø recessed high-contrast cups
- Center Divider: Recessed 1.2mm scoring groove
- Pitch: 14.0 mm half matches Ten-Frame cell

12-FRAME REGROUPING TRAY & DOUBLES LID

Lid features doubles spine [0|0]–[6|6] and a regrouping tray to model part-whole sums!

DOMINO SUBITIZING & SUM: [5|4] + [3|2] = 14



[5|4] = 9 [3|2] = 5

PART-WHOLE ADDITION

[5|4] pips = 9
[3|2] pips = 5
Total: 9 + 5 = 14
T(7) = 7×8/2 = 28 Tiles
Double-Six Combinatorics

2. KIT PARTS INVENTORY (33 PIECES)

COMPONENT	QTY	SPEC / DIMENSION	PALETTE COLOR
Double-Six Tiles	28x	28.0 × 14.0 × 7.0 mm (T7)	Stealth Black
Wildcard Blanks	2x	28.0 × 14.0 × 7.0 mm [|]	Stealth Black
12-Frame Regroup Tray	1x	Dual 6-Bay Tactile Stage	Matte Charcoal
Doubles Organizer Lid	1x	7-Slot Doubles Spine Rail	Matte Charcoal
3x1 Storage Bin	1x	49.0 mm Height (7U)	Matte Charcoal

TOTAL: 30 Domino Tiles + 12-Frame Tray + Doubles Lid + Storage Bin = 33 Parts

HIGH-CONTRAST PIPS & TEXTURE

- Stealth Black Matte: Low glare, fingerprint resistant polymer.
- Brilliant White Pips: Standard dice quincunx configurations.
- Doubles Spine: Labeled slots [0|0] to [6|6] for organizing.
- High Durability: Scratch-resistant tactile surfaces.

3. WHY KEEP THIS REFERENCE CARD?

✓ **NO LOST PIECE LEFT BEHIND**

Classroom manipulatives get lost. Never abandon an incomplete set! Every individual math component is permanently stocked for single replacements.

Scan back QR or visit stackcalc.io/parts
Order individual replacement pieces anytime

✓ **10-SECOND SELF-CHECKING PACK-UP**

Pack-up is a built-in math game! Arrange the 28 standard tiles and 2 wildcards into four neat columns of 7 and one row of 2 in the box floor. A missing tile breaks the smooth rectangular brick! Organizer lid snaps flush for silent zero-rattle carry.

✓ **HANDS-ON TUTORIAL ON REVERSE**

Flip this card over for a complete step-by-step hands-on lesson with concrete numbers, stage arrangements, and calculator cross-references!

Keep this card stored inside the box lid!

STACKCALC ARCHIVAL INSTRUMENT LABS

Designed in Mountain View, California, USA • Non-Toxic BPA-Free.
Classroom Lab Crates (10x / 30x Stations): stackcalc.io/schools

TRIANGULAR NUMBER COMBINATORIAL THEOREM

The number of unordered pairs chosen with replacement from n elements is T(n) = n(n+1)/2.

Universal 7.00mm Datum • Classroom Continuum (10 Kits)

OFFICIAL REFILL PORTAL

stackcalc.io/parts
Order replacement parts anytime

SERIES 1 • CARD 02 OF 10 • DOMAIN: DISCRETE COMBINATORICS & PARTITIONS • 7.00mm 1U PITCH • T(7)=28 TRIANGULAR TILES

LESSON GUIDE: TRIANGULAR COMBINATORICS & PART-WHOLE SUMS

Step-by-Step Hands-On Tutorial: T(7) = 28 Tiles & [5|4] + [3|2] = 14

Grades K-5 Addition & Combinatorics • Hands-On Classroom & Home STEM Studio Series

CARD 02 / 10 • DOMI-01

MATHEMATICAL GUIDE

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN
stackcalc.io/products/math-dominoes/
Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS.MATH.CONTENT.K.OA.A.3 (Decompose to 10)
- CCSS.MATH.CONTENT.1.OA.B.4 (Unknown Addend)
- TEKS 111.3.b.3.A (Model Addition)

Grades K-5 Addition & Combinatorics

💡 **MATHEMATICAL INQUIRY**

If you flip a domino end-for-end, does the total pip sum change? Why or why not?

1 STEP 1: LINE UP DOUBLES SPINE [0|0] TO [6|6]

Place the 7 double tiles in the lid's organizer spine:
[0|0]-0, [1|1]-2, [2|2]-4, [3|3]-6, [4|4]-8, [5|5]-10, [6|6]-12.
These 7 doubles form the diagonal spine of the 28-tile triangle!

2 STEP 2: PAIR COMPLEMENTARY SUM-TO-6 TILES

Find tiles whose pip sum equals 6: [0|6], [1|5], [2|4], [3|3].
Place them in the 12-frame tray side by side.
Observe the inverse symmetry: as left side increases by 1,
the right side decreases by 1, preserving the constant sum 6!

3-4 STEPS 3 & 4: COMBINE TILES & REGROUP

3. COMBINE TWO TILES: Take tile [5|4] (sum 9) and [3|2] (sum 5). Seat in 12-frame tray.
4. CALCULATE SUM: 9 + 5 = 14! (1 full 10-frame + 4 ones).
Calculator Verification: Press Domino Pip Pattern Icon ([g][COUNT] & [g][a:b]) and run 5 [ENTER] 4 [+] 3 [ENTER] 2 [+] [+] = 14.

5 CORE PRINCIPLE: THE 28-TILE TRIANGULAR NUMBER SECRET

Why are there exactly 28 tiles? Count them by their highest pip: 7 tiles have a 0 ([0|0]..[0|6]), 6 new tiles have a 1 ([1|1]..[1|6]), 5 have a 2, down to 1 tile with a 6 ([6|6]).
Sum: 7 + 6 + 5 + 4 + 3 + 2 + 1 = 28! Dominoes are triangular numbers made tactile!

★ **EXTENSION INVESTIGATION: FIND ALL SUM-TO-7 TILES**

Can you find all 3 tiles that sum to 7: [1|6], [2|5], and [3|4]? Why is there no [0|7] tile in a double-six set? (Scan QR for 3D animated solution!)
→ Assemble with your physical blocks, then scan the QR code to verify with the 3D step-by-step vector model.

STACKCALC ARCHIVAL INSTRUMENT LABS

Interactive Mathematics Manipulative Ecosystem.
Designed in Mountain View, California • Archival BPA-Free POM/PETG.

CROSS-REFERENCED CALCULATOR VERIFICATION

Every keystroke sequence verified with RPNCore CalculatorEngine.
Modular Hardware Continuum (9 Kits) • Universal 7.00mm Pitch • stackcalc.io

PARTS & SCHOOL REFILL PORTAL

stackcalc.io/parts
Classroom Lab Crates: stackcalc.io/schools

STACKCALC STEM PEDAGOGY & PHYSICAL COMPUTING

CONCRETE • REPRESENTATIONAL • ABSTRACT

1. TACTILE CONCRETE

7.00mm 1U PITCH
Every physical block dimension is locked to an exact integer ratio, providing tactile feedback of quantity.

MOTOR KINESTHETIC
Sliding, stacking, and interlocking builds spatial memory and intuitive number sense before symbols appear.

CONSERVATION OF QUANTITY
Mass, area, and volume remain invariant regardless of how blocks are grouped or rotated.

SELF-CORRECTING
Precision mechanical detents seat flush only when arithmetic identities and geometric proofs are correct.

CCSS.MATH.PRACTICE.MP5
Use appropriate tools strategically.

2. HARDWARE RPN STACK

POSTFIX ARITHMETIC
[ARG 1] [ENTER] [ARG 2] [OP]. Zero operator ambiguity, zero parentheses, and zero hidden precedence rules.

X, Y, Z, T REGISTERS
Every intermediate result is visible on the display stack. What is stacked on the desk is preserved in memory.

DIGITAL TWIN OF TOWERS
Physically stacking blocks onto a stage lid directly mirrors pushing values into the digital stack registers.

STEP-BY-STEP REASONING
Students observe each operator execute in real time, making compound calculations completely transparent.

CCSS.MATH.PRACTICE.MP1
Make sense of problems & persevere.

3. LEARNING CONTINUUM

GRADES K–2: FOUNDATION
Ten-Frames, Number Rack & Dominoes for subitizing, counting, 5-and-10 anchors, and place value fluency.

GRADES 3–5: FRACTIONS
Fraction bars, prime factorization towers, time gears, and discrete combinatorics with probability dice.

GRADES 6–12: ADVANCED
Algebraic factoring rails, unit circle volvelles, complex numbers, and professional scientific RPN workflows.

DIFFERENTIATED TIERS
Tactile physical manipulates bridge English language learners, visual learners, and accelerated STEM students.

CCSS.MATH.PRACTICE.MP4
Model with mathematics.

CLASSROOM CARE & SANITIZATION

1. FOOD-SAFE RECYCLED POLYMERS:
Manufactured from high-purity virgin polymer (PLA / PETG). Non-toxic, odorless, BPA-free, and compliant with RoHS, EN71-3, and ASTM F963 safety standards.

2. CLASSROOM CLEANING & DISINFECTION:
Wash blocks in warm water with mild liquid dish soap (up to 55°C / 130°F) or wipe with standard classroom disinfectant wipes. Allow to air dry flat before storage.

3. MODULAR SNAPFINITY STORAGE DETENTS:
Stackable 3x1 and 3x3 stage lids feature multi-axis detents tested to 10,000+ opening cycles. Trays pack rattle-free for classroom mobile cart transport.

4. SOCRATIC CLASSROOM DISCUSSION:
Encourage students to physically construct models to test hypotheses before executing calculator calculations: "What invariant property did your physical model prove?"

DOMI-01: IN-BOX SPECIFICATION & QC AUDIT

CALIFORNIA LAB QC

CALIFORNIA LAB QC AUDIT CHECKLIST

[✓] MITUTOYO DIGITAL CALIPER DATUM:
All block dimensions conform to ±0.05 mm tolerance across universal 7.00 mm 1U visible height system.

[✓] 1.85 MM RETENTION SOCKET INTERFERENCE:
Stacking plugs seat flush with 0.15 mm diametral interference for zero-wobble tower verticality.

[✓] SNAPFINITY ZERO-RATTLE DETENT LOCK:
Multi-axis lid detents lock securely over 3x1 bin perimeter rim. Audible snap with 1.4 N retention.

[✓] CONTINUOUS 360° MICRO-FLUTING & SEAM TRAP:
2.0 mm pitch, 0.50 mm flutes. Slicer seam locked into rear-left corner seam trap notch.

[✓] FOOD-SAFE RECYCLED POLYMER:
100% virgin PLA / PETG. RoHS and EN71-3 compliant, non-toxic, odorless, child-safe.

[✓] IN-BOX COLLATERAL PLACEMENT:
Place this reference card inside the box lid or inside the storage tray.

RPN HARDWARE COMPATIBILITY

- Compatible with StackCalc SC-6 (Early STEM)
- Compatible with StackCalc SC-32 (Scientific RPN)
- Zero parentheses, zero operator precedence errors
- Physical towers are the exact digital twin of the stack!

TOTAL KIT INVENTORY: 33 PIECES

Verified by Assembly Tech & Machine Vision at Packing Bench.

Inspection Pass • Mountain View, CA

LIFETIME REPLACEMENT GUARANTEE

NO LOST PIECE LEFT BEHIND™

Classrooms lose blocks. We understand. Every individual manipulative block has a permanent part number. Visit stackcalc.io/parts to order exact replacement pieces with same-day dispatch ($1.99–$2.99/piece).

PORTAL: stackcalc.io/parts

DISTRICT & CLASSROOM PROCUREMENT

- Net-30 Invoicing for US Public & Private School Districts.
- Volume Classroom Bundles (10-packs & 30-packs).
- Free Teacher Activity Books & Lesson Blueprints.
- Title I STEM & Carl D. Perkins Grant eligible.

Email: procurement@stackcalc.io
stackcalc.io/schools

3D INTERACTIVE VIDEO DEMOS

Scan the QR code on the back of this card (Side B) to access:

- 60fps 3D animated walkthrough of each lesson
- Interactive web 3D model with explode view
- Step-by-step calculator keystroke simulations
- Printable student worksheets and answer keys

VIDEOS: stackcalc.io/products

Designed & Engineered in Mountain View, California